

ME594 3D Robotics
HW6 Answer to selected questions

1)

$$\omega = \begin{bmatrix} 1 & 0 & \sin\theta \\ 0 & \cos\phi & -\sin\phi \cos\theta \\ 0 & \sin\phi & \cos\phi \cos\theta \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\omega = A \dot{\Theta}$$

Using the above formula

$$\omega = [0.1742, 0.0362, 0.3510] \text{ (answer)}$$

$$\Omega = \begin{bmatrix} \cos\psi \cos\theta & \sin\psi & 0 \\ -\sin\psi \cos\theta & \cos\psi & 0 \\ \sin\theta & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

$$\Omega = A_{\Omega} \dot{\Theta}$$

$$\Omega = [0.1929, 0.1104, 0.3247] \text{ (answer)}$$

2)

i) Compute the rotation matrix, R

$$R = R_x(\phi) * R_y(\theta) * R_z(\psi)$$

ii) Compute the quaternion from rotation or directly from Euler angles

$$q = [0.5000 \ 0.5000 \ -0.1830 \ 0.6830]$$

iii) quaternion derivative is given by

$$\dot{q} = 0.5 \omega \cdot q$$

Here the left side is a quaternion product. Omega should be written as [0 omega] to ensure that the product is consistent.

$$\dot{q} = [-0.2199 \ -0.1150 \ -0.1604 \ 0.2021] \text{ (answer)}$$

$$\text{Method 1: } \Omega = R^T \omega$$

$$\text{Method 2: } \Omega = 2\bar{q} \cdot \dot{q}$$

Here the conjugate part has the imaginary terms (last 3) negated.

$$\Omega = [-0.040192 \ 0.118301 \ 0.704904] \text{ (answer)}$$

3)

tip_linvel = [-0.55529581 -0.32464043 0.05783236]

tip_angvel = [0.65756279 -1.448484 0.01030098]

lower_forearm_com_linvel = [-0.27699101 -0.19849493 -0.05908374]

lower_forearm_com_angvel = [0.54118076 -1.60818157 -0.4075203]